

JavaScript Splice Method Assignment

Part A: Multiple Choice Questions

1. What does the splice() method return?
a) The modified array
b) An array containing the removed elements
c) The number of elements removed
d) undefined
2. What is the correct syntax for splice()?
a) array.splice(start, end)
b) array.splice(index, element)
c) array.splice(start, deleteCount, item1, item2, ...)
d) array.splice(element, count)
3. What will be the output?

b) 5

```
1 let arr = [1, 2, 3, 4, 5];  
2 arr.splice(2, 0, 'a');  
3 console.log(arr.length);  
4
```

a) 4

c) 6

d) 3

4. To add elements without removing any, what should be the deleteCount parameter? a) -1
b) 0
c) 1
d) null
5. What happens when you use a negative start index in splice()? a) It throws an error
b) It counts from the end of the array
c) It starts from index 0
d) It does nothing

Part B: Short Answer Questions

1. Write the output of the following code:

```
1
  let fruits = ['apple', 'banana', 'orange']; 2
  let removed = fruits.splice(1, 1);
3
  console.log(removed);
4
  console.log(fruits);
5
```

2. Explain the difference between these two splice operations:

```
1
  arr.splice(2, 1);
2
  arr.splice(2, 0, 'new'); 3
```

3. How would you remove the last 3 elements from an array using splice()?

4. Write code to replace 2 elements starting from index 1 with a single element 'replaced'. 5.
What will be the result?

```
1      let nums = [10, 20, 30, 40]; 2  
      nums.splice(4, 0, 50);  
3      console.log(nums);  
4
```

Part C: Code Writing

- Given the array ['red', 'blue', 'green', 'yellow', 'purple'], write code to:
Remove 'green' using splice()
Show the removed element and the modified array
- Create an array of weekdays and insert 'Weekend' between Friday and Saturday using splice().
- Write code to replace the middle 3 elements of this array with 'MIDDLE': [1, 2, 3, 4, 5, 6, 7]
- Use splice() to:
Remove the first 2 elements from an array
Add 3 new elements at the beginning
Do both operations in a single splice() call
- Create a function insertAt(array, index, elements) that uses splice() to insert multiple elements at a specific index.

Part D: Problem Solving

- Shopping Cart Manager:
Create an array with 5 products

Remove the product at index 2

Add 2 new products at that position

Remove the last product

Print the final cart

2. Playlist Editor:

Start with an array of 6 song names

Remove songs from index 2 to 4

Add 3 new songs at index 2

Replace the first song with a new one

Show the playlist after each operation

Answer Sheet

Part A: Multiple Choice Answers

1. **b) An array containing the removed elements**

`splice()` returns an array of removed elements, even if empty

2. **c) `array.splice(start, deleteCount, item1, item2, ...)`**

This is the correct syntax for `splice()`

3. **c) 6**

`splice(2, 0, 'a')` adds 'a' at index 2 without removing anything

4. **b) 0**

`deleteCount` of 0 means no elements will be removed

5. **b) It counts from the end of the array**

Negative indices count backwards from the end

Part B: Short Answer Solutions

1. Output:

```
1      ["banana"]
2      ["apple", 'orange'] 3
```

`splice(1, 1)` removes 1 element at index 1 ('banana')

Returns array of removed elements

2. Difference:

`arr.splice(2, 1)`: Removes 1 element at index 2

`arr.splice(2, 0, 'new')`: Adds 'new' at index 2 without removing anything

3. Remove last 3 elements:

```
1      arr.splice(-3, 3);
2
3      // or
4      arr.splice(arr.length - 3, 3);
```

4. Replace 2 elements:

4

1

2

3

5. Result:

`let arr = ['a', 'b', 'c', 'd'];`

```
arr.splice(1, 2, 'replaced'); // Result: ['a', 'replaced', 'd']
```

```
1 [10, 20, 30, 40, 50]
```

2

splice(4, 0, 50) adds 50 at index 4 (end of array)

Part C: Code Writing Solutions

1. Remove 'green':

```
1
  let colors = ['red', 'blue', 'green', 'yellow', 'purple']; 2
  let removed = colors.splice(2, 1);
3
  console.log('Removed:', removed); // ['green']
4
  console.log('Array:', colors); // ['red', 'blue', 'yellow', 'purple'] 5
```

2. Insert 'Weekend':

```
1
  let weekdays = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']; 2
  weekdays.splice(5, 0, 'Weekend');
3
  console.log(weekdays);
4
```

```
// ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Weekend', 'Saturday', 'Sunday'] 5
```

3. Replace middle 3 elements:

```
1
  let arr = [1, 2, 3, 4, 5, 6, 7];
2
  arr.splice(2, 3, 'MIDDLE');
3
  console.log(arr); // [1, 2, 'MIDDLE', 6, 7] 4
```

4. Remove and add in one call:

```
1
  let arr = ['a', 'b', 'c', 'd', 'e'];
2
  arr.splice(0, 2, 'x', 'y', 'z');
3
  console.log(arr); // ['x', 'y', 'z', 'c', 'd', 'e'] 4
```

5. insertAt function:

```
1
  function insertAt(array, index, elements) { 2
    array.splice(index, 0, ...elements); 3
    return array;
4
  }
5
6
  let arr = [1, 2, 3, 4];
7
```

```
insertAt(arr, 2, ['a', 'b', 'c']);  
8  
console.log(arr); // [1, 2, 'a', 'b', 'c', 3, 4] 9
```

Part D: Problem Solving Solutions

1. Shopping Cart Manager:

```
1  
  let cart = ['laptop', 'mouse', 'keyboard', 'monitor', 'webcam']; 2  
  console.log('Initial cart:', cart);  
3  
4  
  // Remove product at index 2  
5  
  let removed = cart.splice(2, 1);  
6  
  console.log('Removed:', removed);  
7  
8  
  // Add 2 new products at that position  
9  
  cart.splice(2, 0, 'headphones', 'microphone');  
10  
  console.log('After adding:', cart);  
11  
12  
  // Remove last product  
13  
  cart.splice(-1, 1);  
14  
  console.log('Final cart:', cart);
```

15

```
// Final: ['laptop', 'mouse', 'headphones', 'microphone', 'monitor'] 16
```

2. Playlist Editor:

1

```
let playlist = ['Song1', 'Song2', 'Song3', 'Song4', 'Song5', 'Song6']; 2  
console.log('Original:', playlist);
```

3

4

```
// Remove songs from index 2 to 4
```

5

```
playlist.splice(2, 3);
```

6

```
console.log('After removal:', playlist);
```

7

8

```
// Add 3 new songs at index 2
```

9

```
playlist.splice(2, 0, 'NewSong1', 'NewSong2', 'NewSong3'); 10  
console.log('After adding:', playlist);
```

11

12

```
// Replace first song
```

13

```
playlist.splice(0, 1, 'Favorite Song');
```

14

```
console.log('Final playlist:', playlist);
```

15

```
// Final: ['Favorite Song', 'Song2', 'NewSong1', 'NewSong2', 'NewSong3', 'Song6'] 16
```